



UNIVERSITY OF NAIROBI

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DEPARTMENT OF MECHANICAL AND MANUFACTURING ENGINEERING

References Draft

Project: Bridging the Gap: Computational Prediction vs. Physical Performance of 3D-Printed Lattice Structures

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The following references have been selected to provide the theoretical foundation, manufacturing parameters, and testing standards required for the project:

- [1] L. J. Gibson and M. F. Ashby, *Cellular Solids: Structure and Properties*, 2nd ed. Cambridge: Cambridge University Press, 1997.
- [2] R. G. Silva, C. S. Estay, G. M. Pavez, J. Z. Viñuela, and M. J. Torres, “Influence of Geometric and Manufacturing Parameters on the Compressive Behavior of 3D Printed Polymer Lattice Structures,” *Materials*, vol. 14, no. 6, p. 1462, 2021.
- [3] F. Tamburrino, S. Graziosi, and M. Bordegoni, “The design process of additively manufactured mesoscale lattice structures: A review,” *Journal of Computing and Information Science in Engineering*, vol. 18, no. 4, p. 040801, 2018.
- [4] L. B., “Tensile and Compressive Behavior in the Experimental Tests for PLA Specimens Produced via Fused Deposition Modelling Technique,” *Technologies*, vol. 4, no. 3, p. 140, 2020.
- [5] S. Dixit and S. Jain, “Influence of layer thickness, printing speed, and infill density on the compressive strength of FDM-printed TPU- and PLA-based BCC lattice structures,” *Eur. Mech. Sci.*, vol. 7, pp. 278–284, 2023.